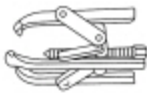
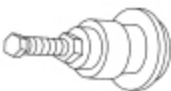


Engine Front Cover

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

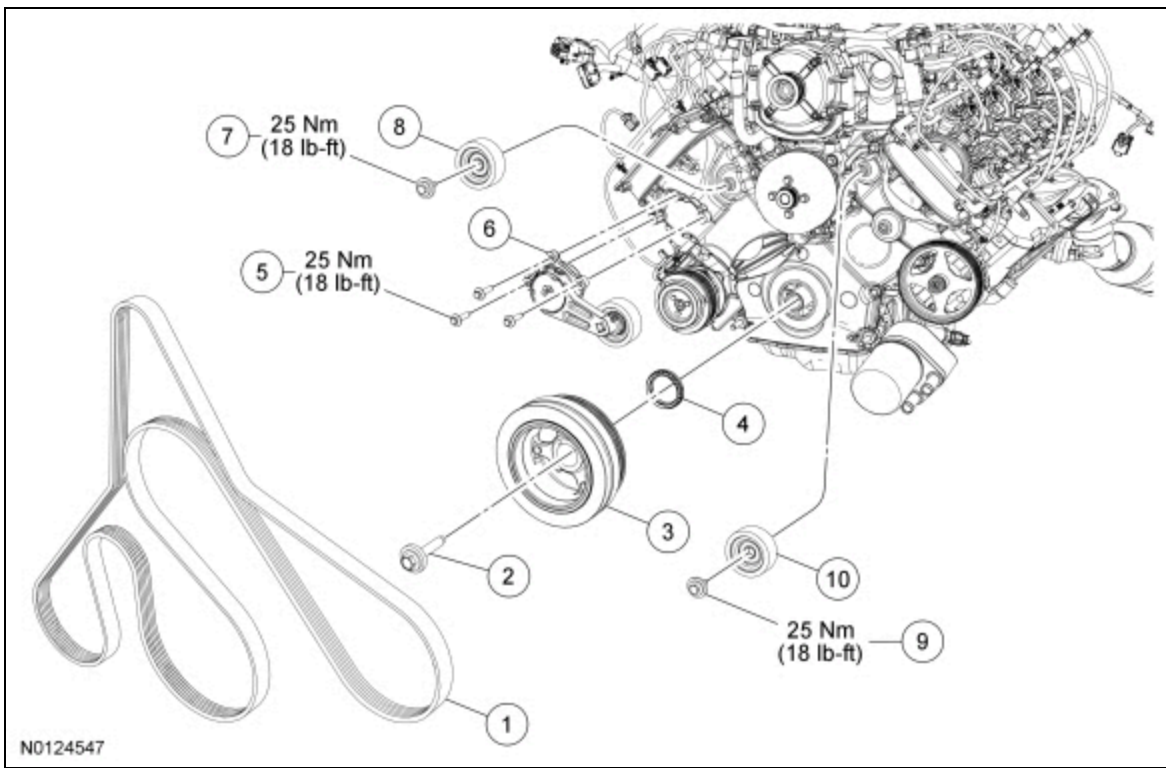
 ST1184-A	3 Jaw Puller or equivalent 303-D121
 ST2132-A	Adapter Set, Step Plate 205-DS011 or equivalent
 ST3245-A	Installer, Crankshaft Damper and Crankshaft Front Seal 303-1508
 ST1328-A	Installer, Front Cover Oil Seal 303-335 (T88T-6701-A)
 ST1385-A	Remover, Oil Seal 303-409 (T92C-6700-CH)
 ST1438-A	Strap Wrench 303-D055 (D85L-6000-A) or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Material

Item	Specification

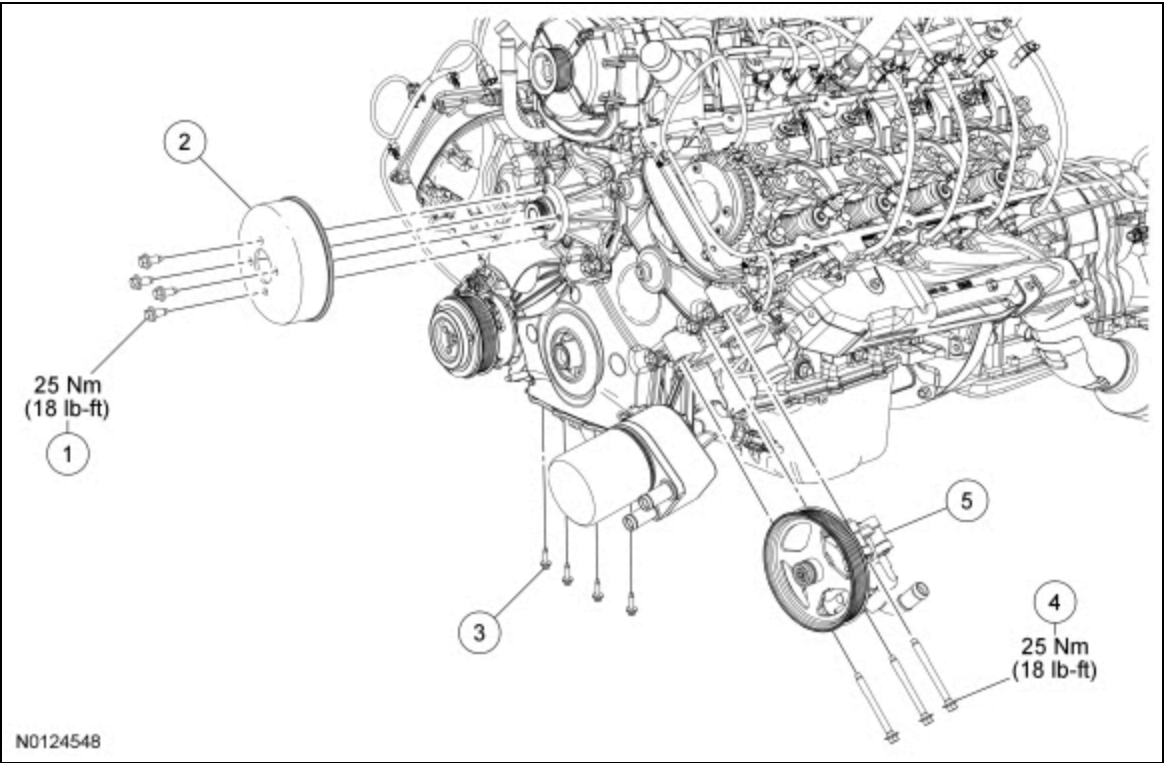
Motorcraft® MERCON® V Automatic Transmission Fluid XT-5-QMC (US); CXT-5-LM12 (Canada)	MERCON® V
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Silicone Gasket and Sealant TA-30	WSE-M4G323-A4

Front End Accessory Drive (FEAD) , Crankshaft Pulley and Crankshaft Front Seal



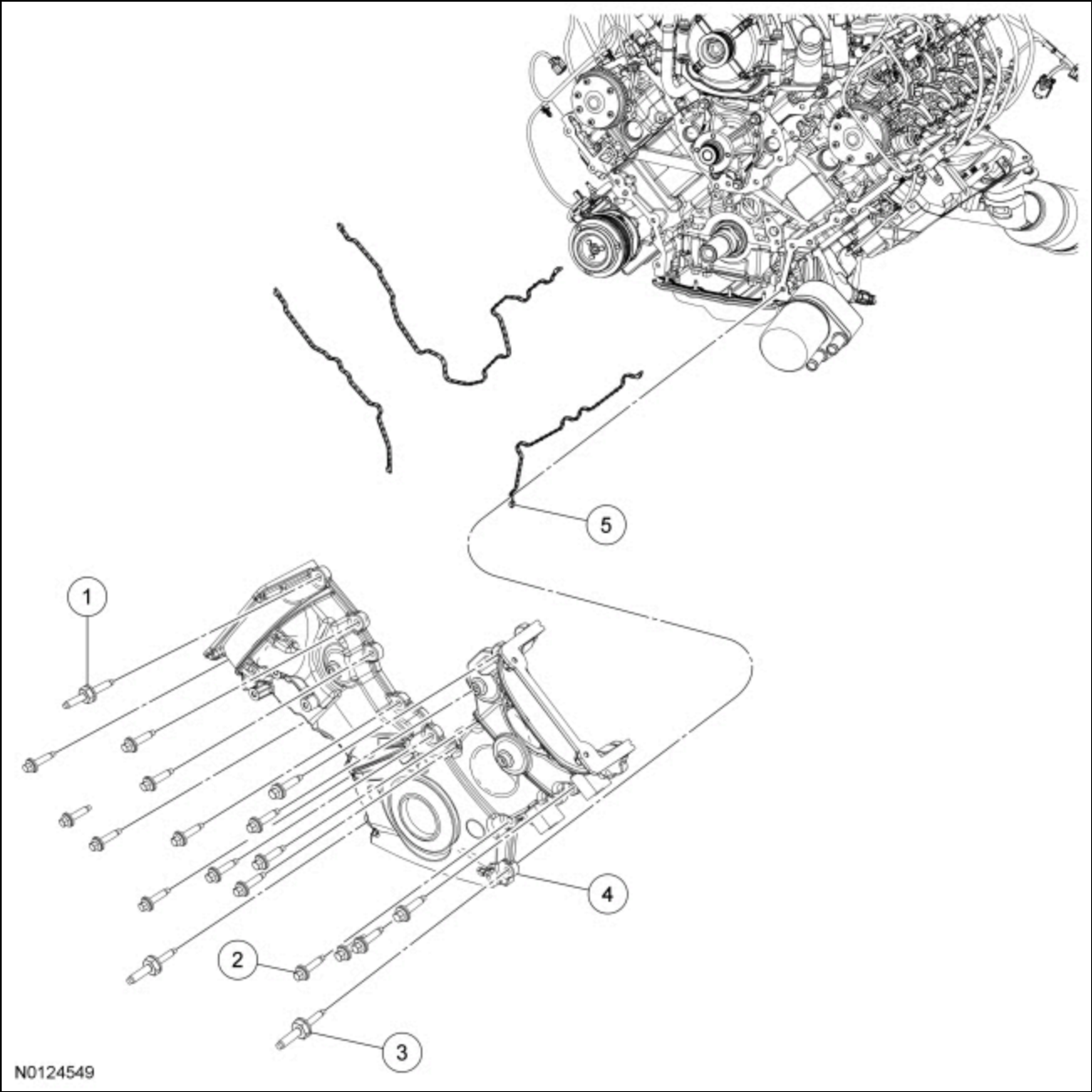
Item	Part Number	Description
1	8620 8620	Accessory drive belt
2	W701512 W701512	Crankshaft pulley bolt
3	6316 6316	Crankshaft pulley
4	6700 6700	Crankshaft front seal
5	N808920 N808920	Accessory drive belt tensioner bolt (3 required)
6	6B209 6B209	Accessory drive belt tensioner
7	N808102 N808102	RH accessory drive belt idler pulley bolt
8	19A216 19A216	RH accessory drive belt idler pulley
9	N808102 N808102	LH accessory drive belt idler pulley bolt
10	19A216 19A216	LH accessory drive belt idler pulley

Coolant Pump Pulley, Power Steering Pump and Front Oil Pan Bolts



Item	Part Number	Description
1	N806282 N806282	Coolant pump pulley bolt (4 required)
2	8A528 8A528	Coolant pump pulley
3	W701605 W701605	Oil pan bolt (4 required)
4	—	Power steering pump bolts (3 required)
5	—	Power steering pump

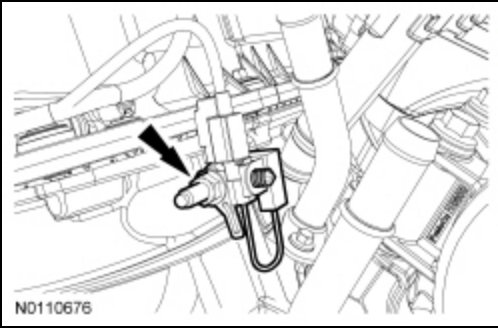
Engine Front Cover and Gaskets



Item	Part Number	Description
1	W713461 W713461	Engine front cover stud
2	W713261 W713261	Engine front cover bolt (17 required)
3	W713462 W713462	Engine front cover stud (2 required)
4	6C086 6C086	Engine front cover
5	—	Engine front cover gasket (part of 6C086) (3 required)

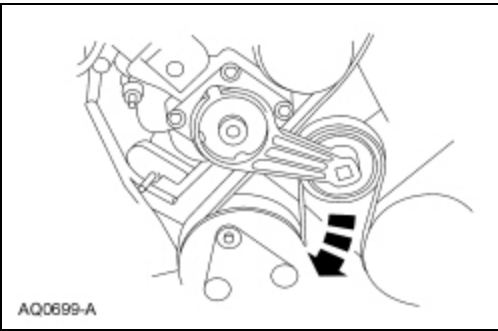
Removal

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. Remove the RH and LH Variable Camshaft Timing (VCT) variable force solenoids. For additional information, refer to Section 303-14.
3. Remove the nut and the position the radio ignition interference capacitor aside.



4. Loosen the 4 coolant pump pulley bolts.

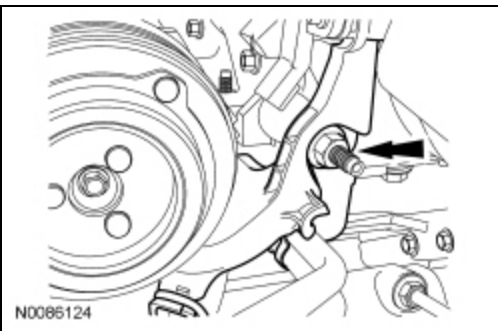
5. Rotate the tensioner clockwise and remove the accessory drive belt.



6. Remove the 4 bolts and the coolant pump pulley.

7. Drain the engine oil.

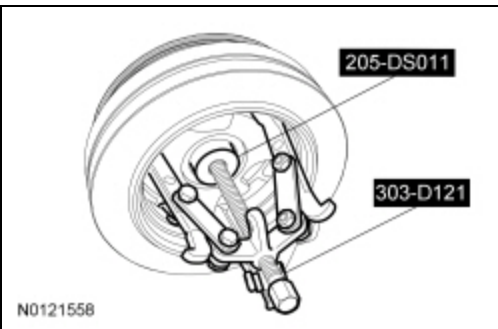
8. Remove the nut and position aside the transmission cooler tube support bracket and the starter wiring harness support bracket.



9. Remove the crankshaft pulley bolt.

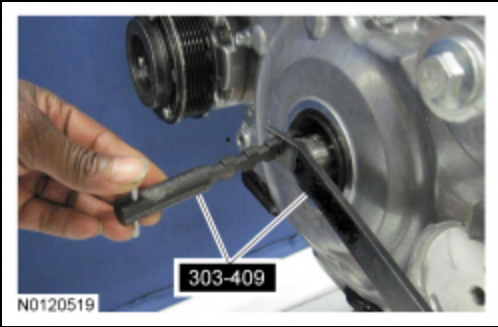
- Discard the crankshaft pulley bolt.

10. Using the 3 Jaw Puller and Step Plate, remove the crankshaft pulley.

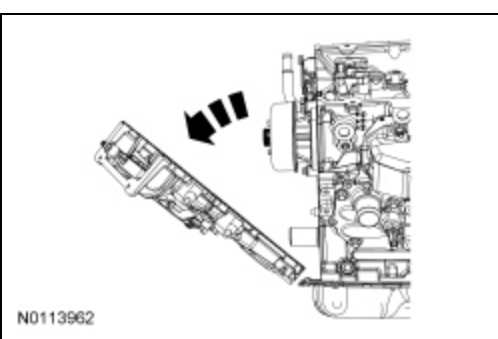


11. **NOTICE:** Use care not to damage the engine front cover or the crankshaft when removing the seal.

Using the Oil Seal Remover, remove the crankshaft front oil seal.



12. Remove the 2 bolts and the 2 accessory drive idler pulleys.
13. Remove the 3 bolts and the accessory drive belt tensioner.
14. Remove the 4 front oil pan bolts.
15. Remove the power steering pump. For additional information, refer to Section 211-02.
16. Remove the 17 bolts and the 3 studs.
17. Remove the engine front cover from the front cover-to-cylinder block dowel.
 - Remove and discard the engine front cover gaskets.



18. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

Clean the mating surfaces with silicone gasket remover and metal surface prep. Follow the directions on the packaging.

- Inspect the mating surfaces.

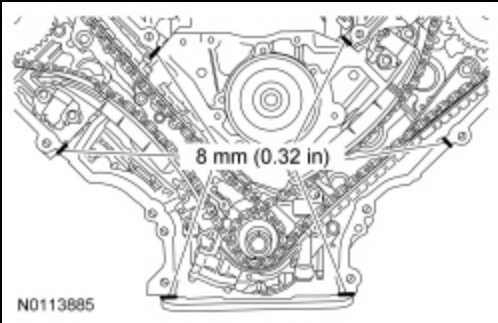
Installation

1. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

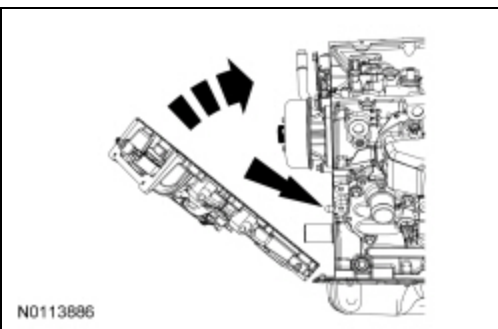
NOTE: *If the engine front cover is not secured within 5 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

NOTE: *Make sure that the engine front cover gasket is in place on the engine front cover before installation.*

Apply a bead of silicone gasket and sealant along the cylinder head-to-cylinder block surface and the oil pan-to-cylinder block surface, at the locations shown.



2. Install new engine front cover gaskets on the engine front cover. Position the engine front cover onto the dowels. Install the fasteners finger-tight.



3. Tighten the 20 engine front cover fasteners in the sequence shown in 2 stages.

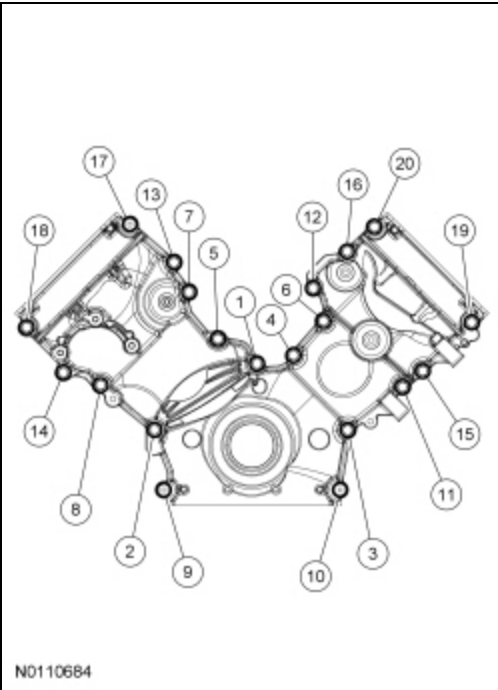
Stage 1: Tighten all fasteners to 10 Nm (89 lb-in).

Stage 2: Tighten all fasteners to 20 Nm (177 lb-in).

Stage 3: Tighten all fasteners an additional 45 degrees.

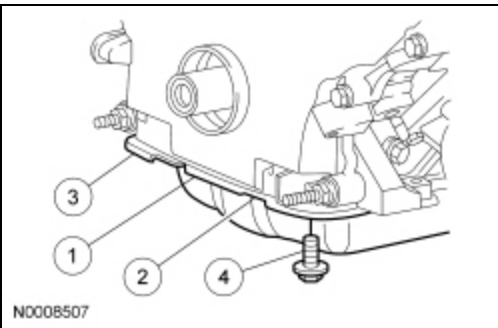
Item	Part Number	Description
1	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
2	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
3	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
4	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
5	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
6	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
7	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
8	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
9	W713462	Stud, Hex Head Pilot, M8 x 33 + M10 x 30
10	W713462	Stud, Hex Head Pilot, M8 x 33 + M10 x 30
11	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
12	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
13	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
14	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
15	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
16	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
17	W713461	Stud, Hex Head Pilot, M8 x 33 + M8 x 27
18	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33
19	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33

Item	Part Number	Description
20	W713261	Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33



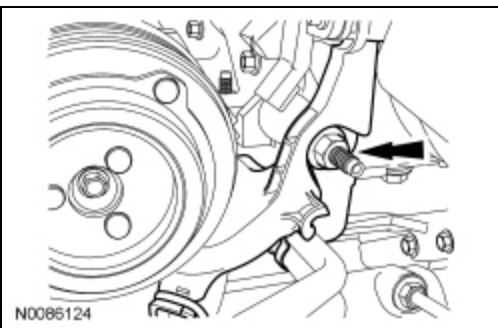
4. Loosely install the 4 bolts, then tighten in 2 stages, in the sequence shown.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.

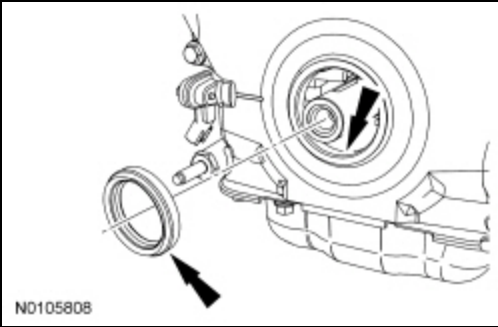


5. Position the starter wiring harness support bracket and the transmission fluid cooler tube support bracket and tighten the nut.

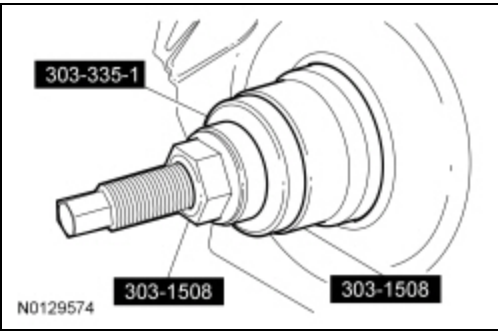
- Tighten to 10 Nm (89 lb-in).



6. Lubricate the engine front cover and the crankshaft seal inner lip with clean engine oil.

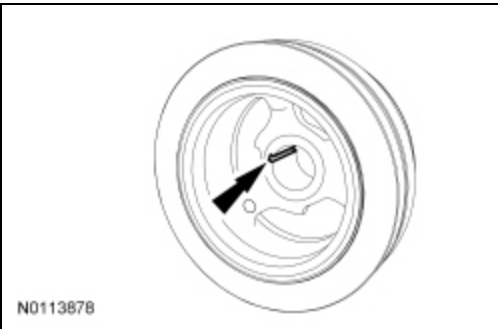


7. Using the Crankshaft Damper and Crankshaft Front Seal Installer and the Front Cover Oil Seal Installer, install a new crankshaft front seal into the engine front cover.

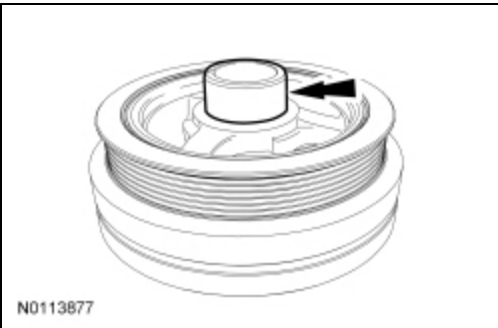


8. **NOTE:** *If not secured within 5 minutes, the sealant must be removed and the sealing area cleaned with metal surface prep and silicone gasket remover. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

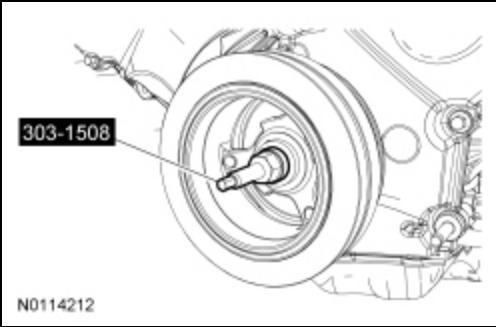
Apply silicone gasket and sealant to the Woodruff key slot in the crankshaft pulley.



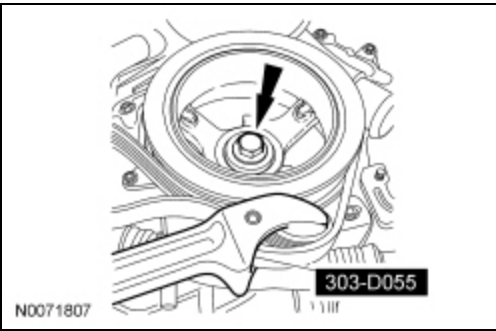
9. Lubricate the crankshaft pulley sealing area with clean engine oil prior to installation.



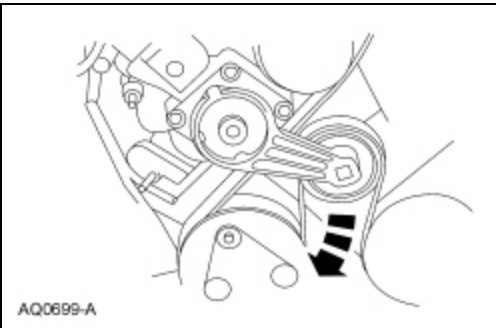
10. Using the Crankshaft Vibration Damper Installer, install the crankshaft pulley.



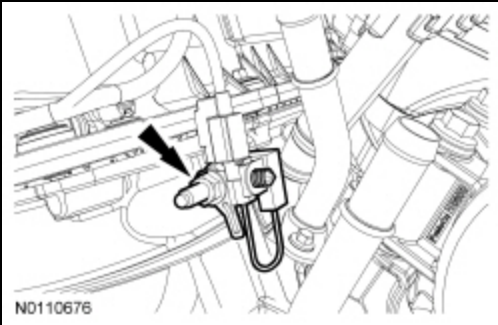
11. Using the Strap Wrench, install a new crankshaft pulley bolt and the original washer, tighten the bolt in 2 stages:
 - Stage 1: Tighten to 175 Nm (129 lb-ft).
 - Stage 2: Tighten an additional 90 degrees.



12. Install the accessory drive belt tensioner and the 3 bolts.
 - Tighten to 25 Nm (18 lb-ft).
13. Install the 2 accessory drive idler pulleys and the 2 bolts.
 - Tighten to 25 Nm (18 lb-ft).
14. Position the coolant pump pulley and install the 4 bolts finger-tight.
15. Install the power steering pump. For additional information, refer to Section 211-02.
16. Rotate the tensioner clockwise and install the accessory drive belt.



17. Tighten the 4 coolant pump pulley bolts.
 - Tighten to 25 Nm (18 lb-ft).
18. Position the radio ignition interference capacitor and install the nut.
 - Tighten to 25 Nm (18 lb-ft).



19. Install the RH and LH ~~VCT~~ variable force solenoids. For additional information, refer to Section 303-14.

20. Fill the crankcase with clean engine oil.